

12. Regression Discontinuity Designs

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Jaewon (“Jay-one”) Yoo

National Tsing Hua University

Outline

1. Sharp Regression Discontinuity Designs
2. Estimation in the SRD
3. Fuzzy Regression Discontinuity Designs
4. R Codes for SRD and FRD

Where are we? Where are we going?

- So far:
 - Randomized experiments identify causal effects.
 - Regression, matching, weighting, DML for selection on observables.
 - Instrumental variables for when this doesn't hold.
 - Panel methods (e.g., DiD, synthetic control, FE estimators) when units are observed repeatedly.
- Basic idea: find exogenous variation in the treatment assignment.
 - RCT: randomization provides exogenous variation.
 - Selection on observables: treatment as-if random conditional on $\mathbf{X}_i$.
 - IV: Instrument provides exogenous variation.
 - DiD: the $D_i \cdot \text{Post}_t$ interaction (the contrast between treated and control pre/post changes) provides exogenous variation under parallel trends.
- Regression discontinuity: a discontinuity in treatment assignment.



Source: *Chapter 6 of Mostly Harmless Econometrics (Textbook 1)* by J. Angrist & J. Pischke

1/ Sharp Regression Discontinuity Designs

The Setup

- The basic idea behind RDDs:
 - Treatment assignment is determined by a cutoff in some variable, X_i .
 - X_i is a **forcing/running variable**
- Treatment changes discontinuously at the cutoff,
 - but, unobserved confounders vary smoothly around the cutoff.
- $\rightsquigarrow$ changes in the outcome at a threshold have a causal interpretation.
- Classic examples of the setup is in the educational context:
 - Merit scholarships that are allocated based on a test score threshold (Thistlethwaite & Campbell, 1960)
 - Class size on test scores using total student thresholds to create new classes (Angrist & Lavy, 1999)

Sharp RD

- Notations:
 - Treatment: $D_i = 1$ or $D_i = 0$
 - Potential outcomes: $Y_i(1)$ and $Y_i(0)$
 - Observed outcomes: $Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$
 - Continuous forcing variable: $X_i \in \mathbb{R}$ (discrete more complicated)
- **Sharp RD:** $D_i = 1\{X_i \geq c\} \quad \forall i$
 - Treatment is a **deterministic** function of the forcing variable and the threshold.
 - When test scores are above 1,500 $\rightarrow$ offered scholarship
 - When test scores are below 1,500 $\rightarrow$ not offered scholarship
- Note: positivity/overlap violated by design here
 - $\mathbb{P}[D_i = 1 | X_i = c - \varepsilon] = 0$
 - $\mathbb{P}[D_i = 1 | X_i = c + \varepsilon] = 1$
 - $\rightsquigarrow$ Can't use standard identification toolkit for ATE/ATT.

Plotting the RDD (Imbens and Lemieux, 2008)

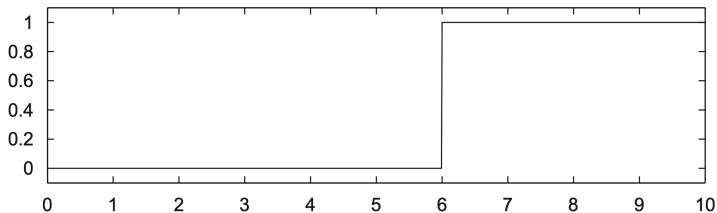


Fig. 1. Assignment probabilities (SRD).

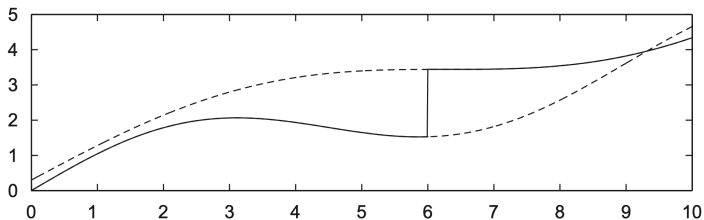


Fig. 2. Potential and observed outcome regression functions.

Source: Figure 4 in Imbens, Guido W., and Thomas Lemieux. "Regression discontinuity designs: A guide to practice." *Journal of econometrics* 142, no. 2 (2008): 615-635.

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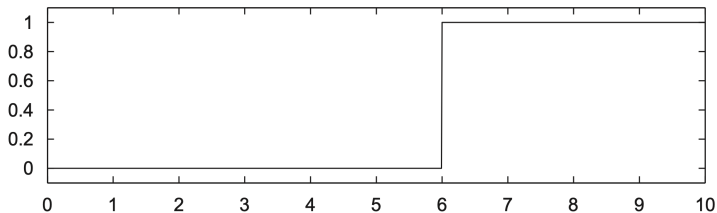


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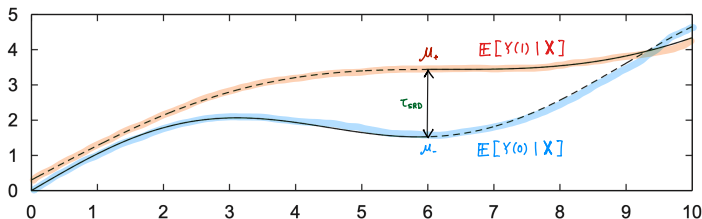


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Quantity of Interest

- Estimand: **local** average treatment effect at the cutoff

$$\begin{aligned}\tau_{\text{SRD}} &= \mathbb{E}[Y_i(1) - Y_i(0) | X_i = c] \\ &= \mathbb{E}[Y_i(1) | X_i = c] - \mathbb{E}[Y_i(0) | X_i = c]\end{aligned}$$

- Very difficult to extrapolate beyond this.
- Problem: X_i is continuous so we never observe $X_i = c$.
 - $\rightsquigarrow$ Identification comes from **extrapolation** around c
 - Extrapolation requires **smoothness**

Continuity of the CEFs

- **Assumption:** CEFs of potential outcomes are **continuous** in X_i
 - $\mu_1(x) = \mathbb{E}[Y_i(1)|X_i = x]$ is continuous
 - $\mu_0(x) = \mathbb{E}[Y_i(0)|X_i = x]$ is continuous
- This continuity implies the following:

$$\begin{aligned}\mathbb{E}[Y_i(0)|X_i = c] &= \lim_{x \uparrow c} \mathbb{E}[Y_i(0)|X_i = x] \quad (\text{continuity}) \\ &= \lim_{x \uparrow c} \mathbb{E}[Y_i(0)|D_i = 0, X_i = x] \quad (\text{SRD}) \\ &= \lim_{x \uparrow c} \mathbb{E}[Y_i|X_i = x] \quad (\text{consistency/SRD})\end{aligned}$$

- Note that this is the same for the treated group:

$$\mathbb{E}[Y_i(1)|X_i = c] = \lim_{x \downarrow c} \mathbb{E}[Y_i|X_i = x]$$

Identification Results

- Consistency + SRD + Continuity $\rightsquigarrow$ Identification:

$$\begin{aligned}\tau_{\text{SRD}} &= \mathbb{E}[Y_i(1) - Y_i(0) | X_i = c] \\ &= \mathbb{E}[Y_i(1) | X_i = c] - \mathbb{E}[Y_i(0) | X_i = c] \\ &= \lim_{x \downarrow c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = x]\end{aligned}$$

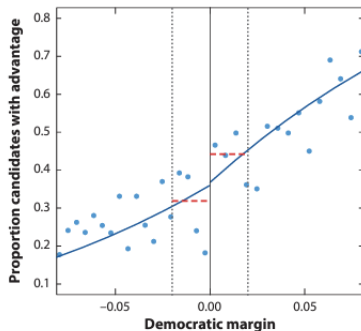
- Problem: Estimate two regression functions at a point.
 - Without parametric assumptions, can be challenging!
 - Nonparametric regression can be consistent, but convergence is slow and
- NB/note well: Not equivalent to **local randomization**,

$$\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp \mathbf{1}\{X_i > c\} \mid c_0 \leq X_i \leq c_1$$

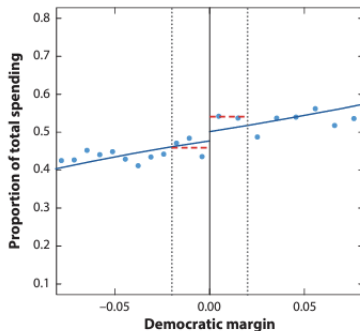
- LR is stronger than continuity b/c it rules out confounding around c .
- Implies no slope in $\mathbb{E}[Y_i(d) \mid X_i = x]$ around c .

Issues with Local Randomization Assumptions

a Democratic experience advantage



b Share of total spending by Democratic candidate



Source: Figure 1 in De la Cuesta, Brandon, and Kosuke Imai. "Misunderstandings about the regression discontinuity design in the study of close elections." *Annual Review of Political Science* 19 (2016): 375-396.

What Can Go Wrong?

- Key question: why is there a discontinuity in D_i but not $Y_i(d)$?
 - What else might change at the cutoff?
 - Using 16 age cutoff for RDD of Korea's game shutdown law?
- **Sorting around the threshold:** possible violation of smoothness.
 - Students retaking exams to pass some threshold for financial aid.
 - Students with more money $\rightsquigarrow$ more exam retaking $\rightsquigarrow$ sorting.

2/ Estimation in the SRD

Bin Plots

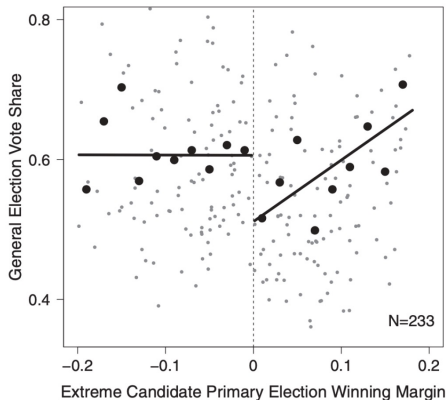
- Standard procedure: **binned means plot** for graphical analysis

$$\bar{Y}_k = \frac{1}{n_k} \sum_{i=1}^N Y_i \cdot \mathbf{1}(b_k < X_i \leq b_{k+1})$$

- where b_k are the bin cutpoints.
- n_k is the number of units within bin k .
- What to observe:
 - Obvious discontinuity at the threshold?
 - Also, are there other unexplained discontinuities?
- Very difficult to sell an RDD without visually obvious results:
 - Imbens & Lemieux (2008): “statistical analysis are just fancy versions of this plot”
 - If it's not in the binned means plot, unlikely to be a robust/credible effect.

Example of a Binned Means Plot

FIGURE 2. General-Election Vote Share After Close Primary Elections Between Moderates and Extremists: U.S. House, 1980–2010



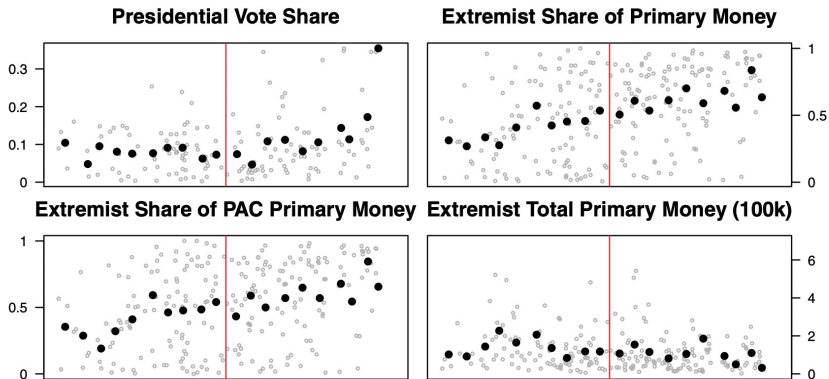
Source: Figure 2 in Hall, Andrew B. "What happens when extremists win primaries?" *American Political Science Review* 109, no. 1 (2015): 18-42.

Other Graphs to Include

- Also good to include binned mean plots for pretreatment covariates.
- Intuition: key assumption in smoothness in the mean of $Y_i(d)$ in X_i .
- Discontinuities in mean of covariates $\rightsquigarrow$ problematic
 - Covariates unaffected by treatment so might indicate sorting.
 - May be an indication of discontinuities in the potential outcome means.
 - Similar to balance tests in matching
- **McCrary test:** plot density of the forcing variable.
 - Separate densities on either side of the cutoff.
 - If there's a discontinuity in the density, maybe a sign of sorting.

Checking Covariates at the Discontinuity

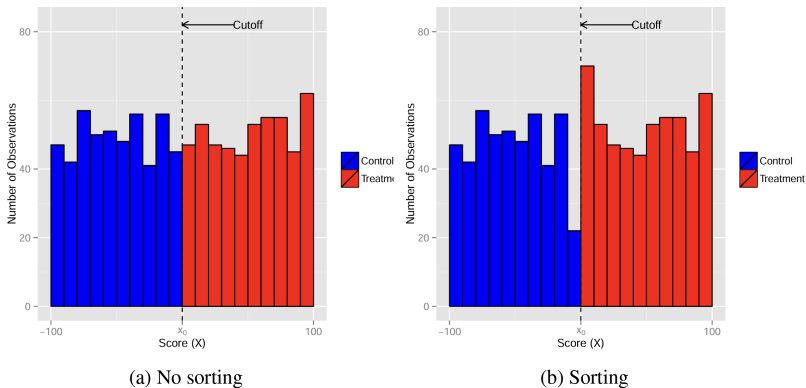
FIGURE A.2. Graphical Balance Tests



Source: *Figure A.2* of Hall, Andrew B. "What happens when extremists win primaries?" *American Political Science Review* 109, no. 1 (2015): 18-42.

McCrary Test

Figure 10: Histogram of Score



Source: Cattaneo, Matias D., Nicolás Idrobo, and Rocío Titiunik (2020). *A Practical Introduction to Regression Discontinuity Designs: Foundations*, Cambridge Elements in Quantitative and Computational Methods for the Social Sciences. Cambridge University Press.

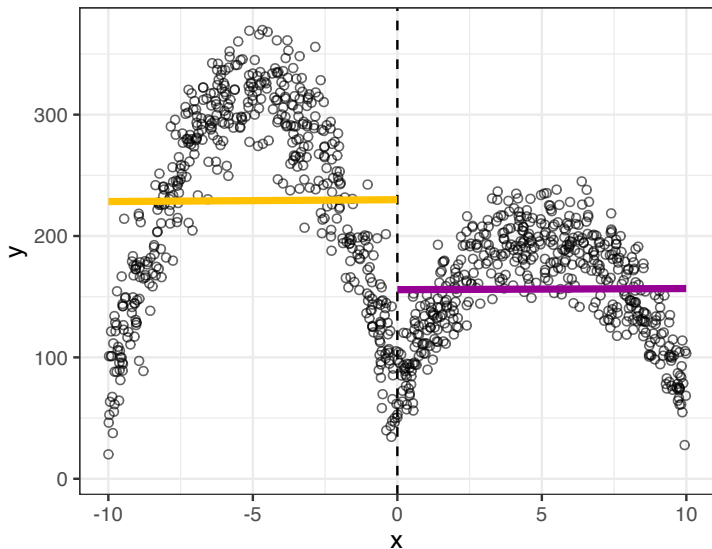
General Estimation Strategy

- The main goal of RD is to estimate the limits of CEFs such as:

$$\lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = x]$$

- Two features different from standard nonparametric regression:
 - We want to estimate this regression at a single point.
 - This point is a **boundary point**, making estimation challenging.
- Bias of nonparametric estimation at a boundary shrinks slowly.
 - Only getting data from one side of the boundary!
- Naive approach: difference in means
 - Problem: uses data too far from the boundary.

Example of Misleading Trends



Nonparametric and Semiparametric Approaches

- Upper and lower limit functions:

$$\mu_+(x) = \lim_{z \downarrow x} \mathbb{E}[Y_i(1) \mid X_i = z]$$

$$\mu_-(x) = \lim_{z \uparrow x} \mathbb{E}[Y_i(0) \mid X_i = z]$$

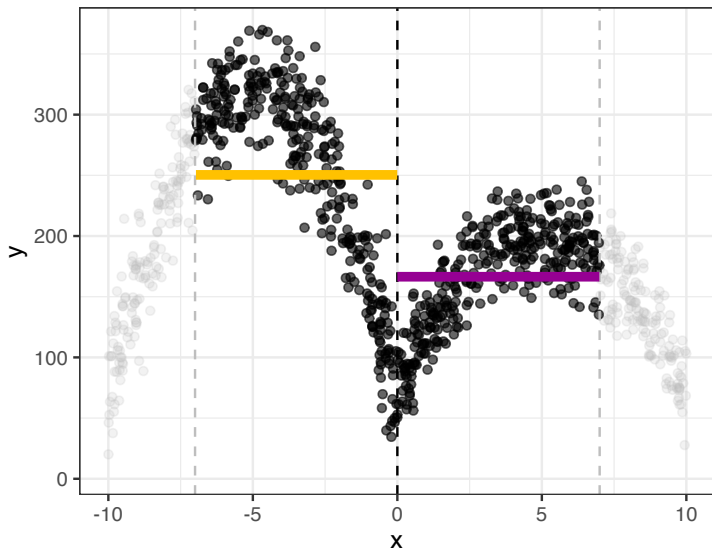
- For the SRD, we have $\tau_{\text{SRD}} = \mu_+(c) - \mu_-(c)$.

- Kernel regression with **uniform kernel**:

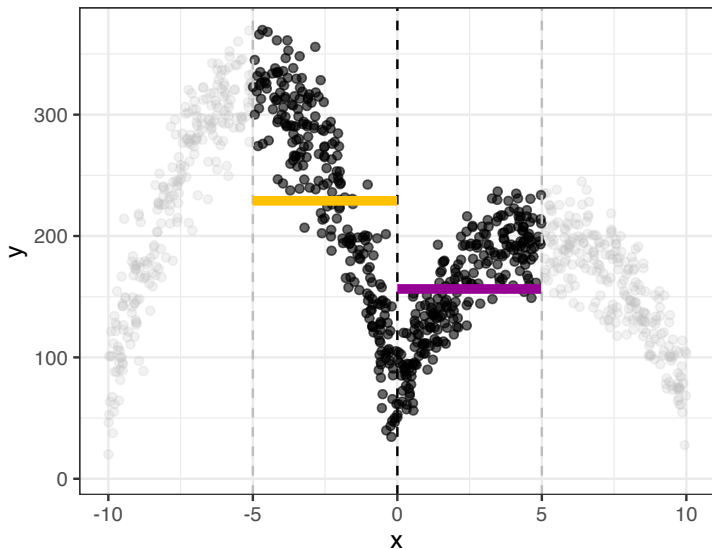
$$\hat{\mu}_-(c) = \frac{\sum_{i=1}^N Y_i \cdot \mathbf{1}\{c-h \leq X_i < c\}}{\sum_{i=1}^N \mathbf{1}\{c-h \leq X_i < c\}}$$

- h is a bandwidth/tuning parameter, selected by you.
- Basically means among units no more than h away from the threshold.

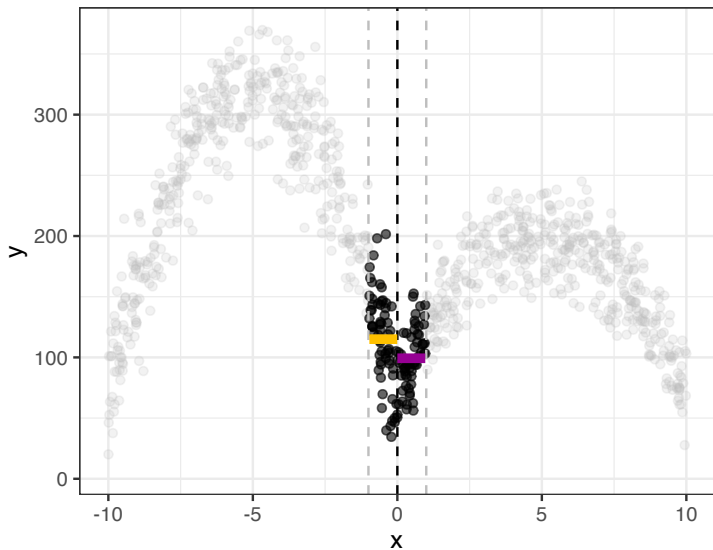
Bandwidth Equal to 7



Bandwidth Equal to 5



Bandwidth Equal to 1



Local Averages

- Estimate mean of Y_i when $X_i \in [c, c + h]$ and when $X_i \in [c - h, c)$.
- Can also view as regression on those units less than h away from c :

$$(\hat{\alpha}, \hat{\tau}_{\text{SRD}}) = \arg \min_{\alpha, \tau} \sum_{i: X_i \in [c-h, c+h]} (Y_i - \alpha - \tau D_i)^2$$

- Predictions about Y_i are locally constant on either side of the cutoff.
- h is a **tuning parameter** that controls the **bias-variance tradeoff**:
 - High h : high bias, low variance (larger n , but farther from the cutoff)
 - Low h : low bias, high variance (smaller n , but closer to the cutoff)
- Downside with averages: bias shrinks slowly as h shrinks.
 - Likely large finite sample bias, poor coverage of confidence intervals.

Local Linear Regression

- Instead of a local constant, we can use a **local linear regression**
- Run a linear regression of Y_i on $X_i - c$ in the group $X_i \in [c - h, c)$:

$$(\hat{\alpha}_-, \hat{\beta}_-) = \arg \min_{\alpha, \beta} \sum_{i: X_i \in [c-h, c)} (Y_i - \alpha - \beta(X_i - c))^2$$

- Same regression for group with $X_i \in [c, c + h]$:

$$(\hat{\alpha}_+, \hat{\beta}_+) = \arg \min_{\alpha, \beta} \sum_{i: X_i \in [c, c+h]} (Y_i - \alpha - \beta(X_i - c))^2$$

- Our estimate is

$$\begin{aligned}\hat{\tau}_{\text{SRD}} &= \hat{\mu}_+(c) - \hat{\mu}_-(c) \\ &= \hat{\alpha}_+ + \hat{\beta}_+(c - c) - \hat{\alpha}_- - \hat{\beta}_-(c - c) \\ &= \hat{\alpha}_+ - \hat{\alpha}_-\end{aligned}$$

More Practical Estimation

- Simplest to use one regression:

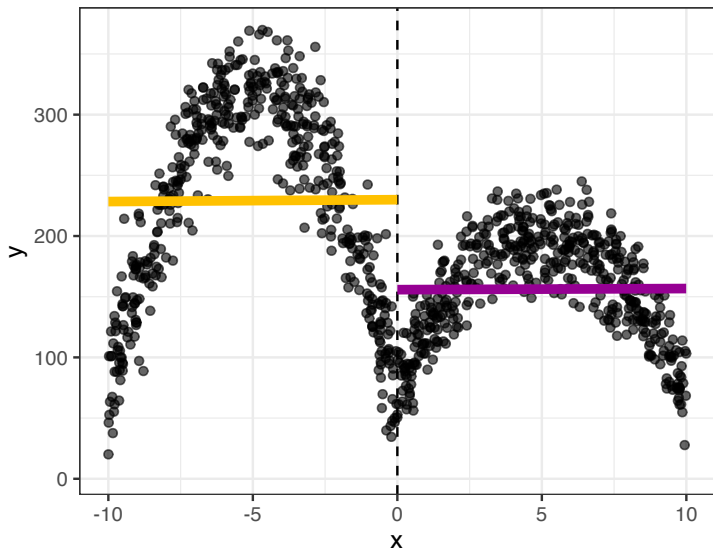
$$\arg \min_{(\alpha, \beta, \tau, \gamma)} \sum_{i: X_i \in [c-h, c+h]} \{Y_i - \alpha - \beta(X_i - c) - \tau D_i - \gamma(X_i - c)D_i\}^2$$

- $\widehat{\tau}_{\text{SRD}} = \widehat{\tau}$ is the coefficient on the treatment.
- Key: interaction between treatment and forcing variable.
- Yields numerically the same as the separate regressions.
- Often better to use a **kernel** to weight points close to c more heavily.

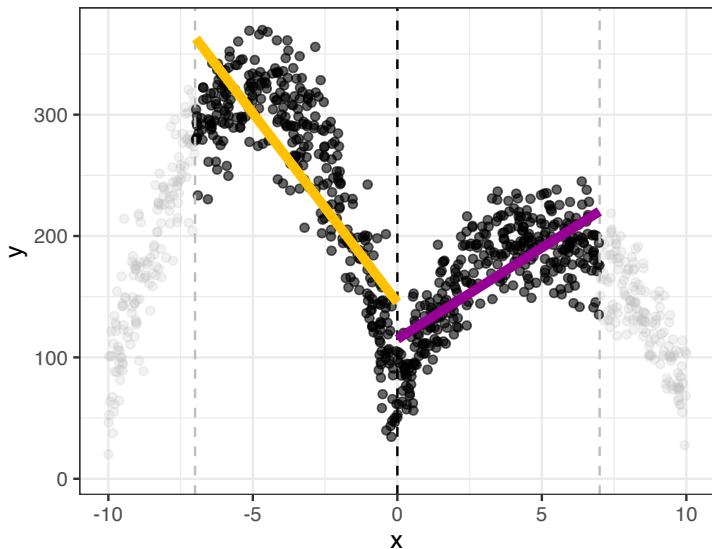
$$\arg \min_{(\alpha, \beta, \tau, \gamma)} \sum_{i=1}^n K\left(\frac{X_i - c}{h}\right) \{Y_i - \alpha - \beta(X_i - c) - \tau D_i - \gamma(X_i - c)D_i\}^2$$

- Popular choice is the **triangular kernel**: $K(u) = (1 - |u|) \cdot \mathbf{1}(|u| < 1)$
- u is standardized distance from the cutoff

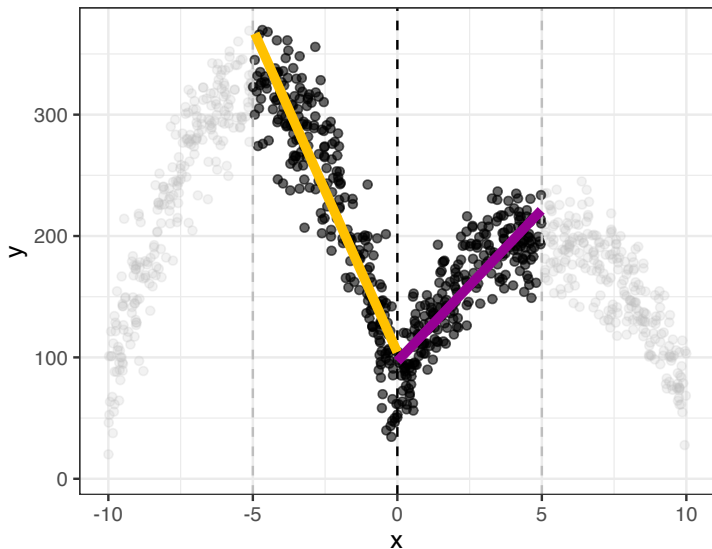
Bandwidth Equal to 10 (Global)



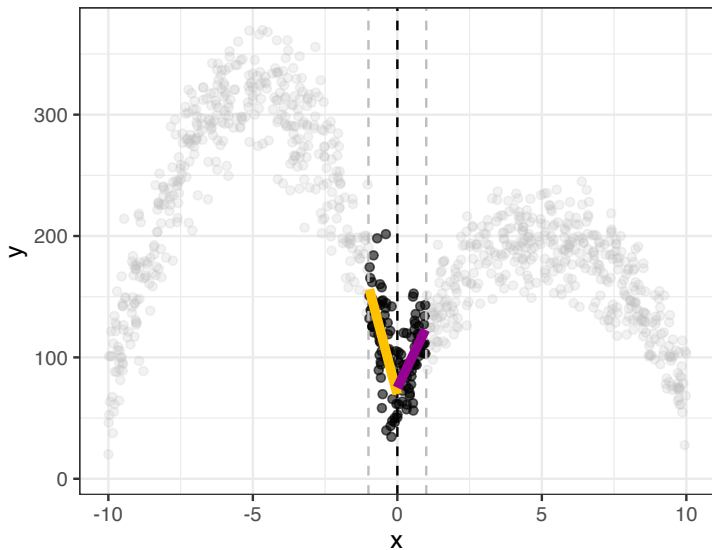
Bandwidth Equal to 7



Bandwidth Equal to 5



Bandwidth Equal to 1



Bandwidths and bias

- Optimal bandwidth shrinks fast enough so $h_n \propto n^{-1/5}$.
 - But this results in asymptotic bias, two possible solutions.
- **Undersmoothing:** have bandwidth shrink more quickly e.g. $h_n \propto n^{-1/4}$
 - Smaller bandwidths $\leadsto$ less bias.
 - Problem: most ways of actually selecting the optimal bandwidth will be too big. Bias strikes back.
- **Robust bias correction:** $\widehat{\tau}_{\text{SRD}}^{\text{rbc}} = \widehat{\tau}_{\text{SRD}} - \widehat{\text{bias}}$
 - Calonico, Cattaneo, and Titiunik (CCT, 2014, Econometrica) gives the form.
 - Allows the use of optimal bandwidths, but need to account for estimation of bias.
 - Bias estimation comes from using higher order polynomials regression.
- Coverage of CIs can be very bad without RBC!

Selecting the optimal bandwidth

- Let $\mathcal{B}$ and $\mathcal{V}$ be approximations of the bias and variance of $\widehat{\tau}_{\text{SRD}}(h)$
 - Based on quadratic approximation of $\mu_d(x)$ rather than linear.
- Idea: find the bandwidth that minimizes the estimation error.

$$MSE(h) = \mathbb{E}[(\widehat{\tau}(h) - \tau_{\text{SRD}})^2 \mid X_1, \dots, X_n] \approx h^4 \mathcal{B}^2 + \frac{1}{nh} \mathcal{V}$$

- Optimal bandwidth: $h_{\text{MSE}} = \left(\frac{\mathcal{V}}{4\mathcal{B}^2} \right)^{1/5} n^{-1/5}$
 - But these depend on unknown biases/variances.
- Procedure:
 1. Pick initial bandwidths to estimate $\mathcal{B}$ and $\mathcal{V}$ with local quadratic regression.
 2. Pick optimal bandwidth for bias correction term and estimate bias with local quadratic regression.
 3. Use both steps to pick optimal bandwidth for local linear regression (h_n)

Odds and ends for the SRD

- **Standard errors:** robust standard errors from local OLS are valid.
 - Not great in finite samples because the bandwidth isn't designed for this purpose.
 - CCT derives nearest neighbors variance estimator that has better coverage.
 - If using RBC, you need to account for that in variance.
- **Covariates:** can add them to the local linear model, but be wary.
 - If covariates are continuous at the cutoff, won't affect estimates much.
 - If they aren't, raises suspicions about identification.
 - ALWAYS REPORT MODELS WITHOUT COVARIATES FIRST
- Possible to use local polynomial regression beyond linear, but performance is poor (very sensitive to end points)
- Use `{rdrobust}` package for CCT bandwidths/estimation.

3/ Fuzzy Regression Discontinuity Designs

The Setup

- Recall treatment: $D_i = 1$ or $D_i = 0$ and forcing variable: X_i .
- **Fuzzy RD**: discontinuity in the probability of treatment.

$$\lim_{x \uparrow c} \mathbb{P}[D_i = 1 \mid X_i = x] \neq \lim_{x \downarrow c} \mathbb{P}[D_i = 1 \mid X_i = x]$$

- No longer deterministic function of forcing variable.
 - SRD is a special case of the FRD.
- Common use case: threshold allows participation in program.
 - Some might not participate even if allowed (noncompliance).
- Forcing variable is an **instrument**:
 - Affects Y_i , but only through D_i (at the threshold).

Fuzzy RD in a Graph (Imbens and Lemieux, 2008)

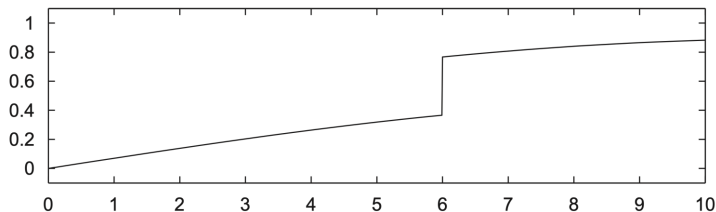


Fig. 3. Assignment probabilities (FRD).

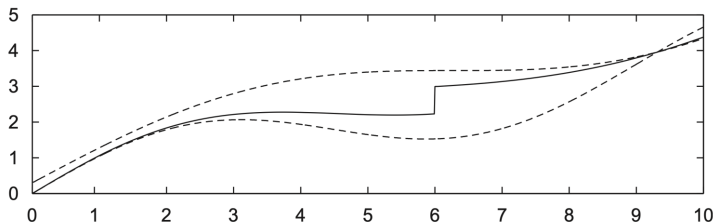


Fig. 4. Potential and observed outcome regression (FRD).

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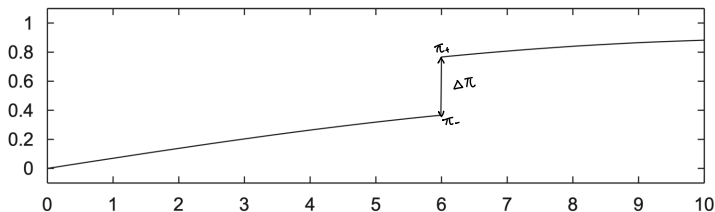


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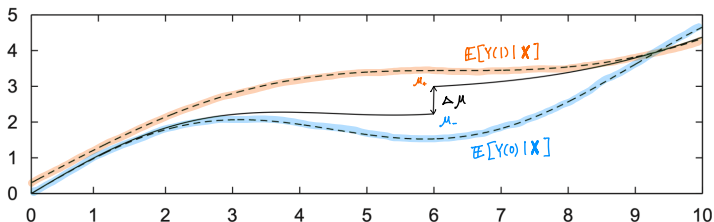


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Fuzzy RD Assumptions

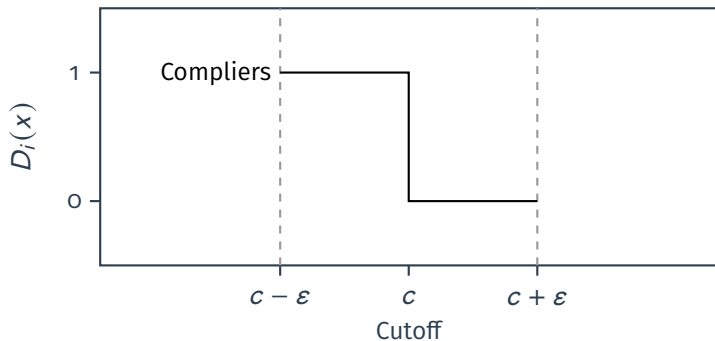
- $D_i(x)$ is potential value of treatment as cutoff changes around c .
 - $D_i(x) = 1$ if unit i would take treatment if cutoff were x .
 - $D_i(x) = 0$ if unit i would take control if cutoff were x .
- **Monotonicity assumption:** $D_i(x)$ is non-increasing in x .
 - Lowering the cutoff can only increase participation.

- Compliers are those i such that for all $0 < e < \varepsilon$:

$$D_i(c - e) = 1 \quad \text{and} \quad D_i(c + e) = 0$$

- Lowering or increasing the threshold would affect their treatment status.
 - Compliance status unobservable
 - $\rightsquigarrow$ Principal strata (Frangakis and Rubin, 2002. Biometrics).
- Example: college students that get above a certain GPA are encouraged to apply to grad school.
 - Compliers wouldn't apply if threshold were slightly higher.
 - Compliers would apply if the threshold were slightly lower.

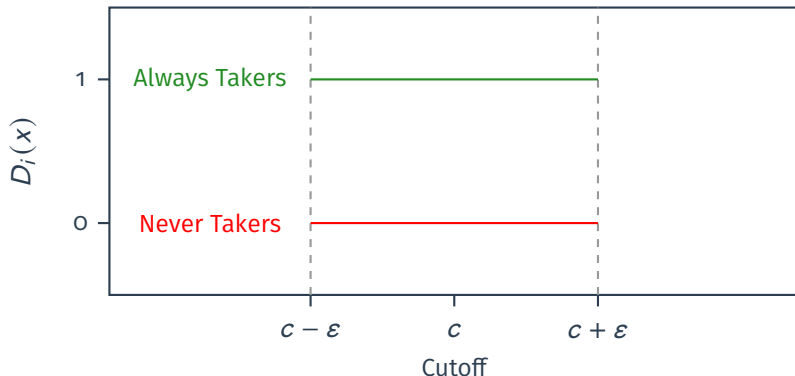
Compliance Graph



- Compliers would not take the treatment if they had $X_i = c$ and we increased the cutoff by some small amount.
- These are compliers at the threshold.

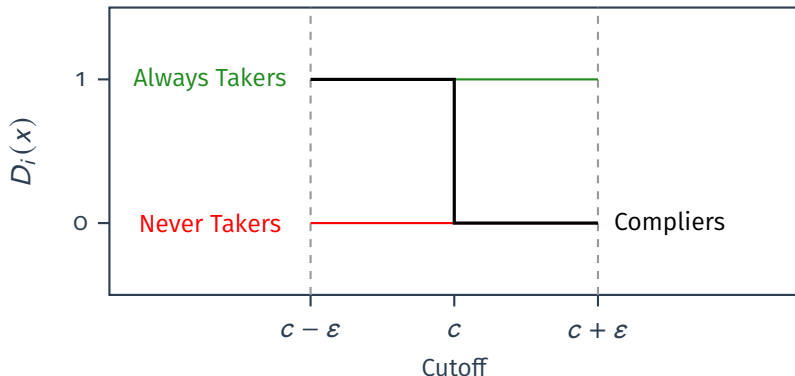
Compliance Groups

- Compliers: $D_i(c - e) = 1$ and $D_i(c + e) = 0$
- Always-takers: $D_i(c + e) = D_i(c - e) = 1$
- Never-takers: $D_i(c + e) = D_i(c - e) = 0$



Compliance Groups

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LATE in the Fuzzy RD

- We can define an estimator that is in the spirit of IV:

$$\begin{aligned}\tau_{\text{FRD}} &= \frac{\lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]}{\lim_{x \downarrow c} \mathbb{E}[D_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[D_i \mid X_i = x]} \\ &= \frac{\text{effect of threshold on } Y_i}{\text{effect of threshold on } D_i}\end{aligned}$$

- Under the FRD assumptions (continuity, consistency, and monotonicity), we can write that the estimator is equal to the effect at the threshold for compliers.

$$\tau_{\text{FRD}} = \mathbb{E}[\tau_i \mid i \text{ is a complier, } X_i = c]$$

- Proof is very similar to the LATE proof.
- External validity? Doubly local $\rightsquigarrow$ careful about generalizing.

Estimation in FRD

- Remember that we had:

$$\tau_{\text{FRD}} = \frac{\lim_{x \downarrow c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = x]}{\lim_{x \downarrow c} \mathbb{E}[D_i | X_i = x] - \lim_{x \uparrow c} \mathbb{E}[D_i | X_i = x]}$$

- Ratio of SRD estimands: use local linear regression for both.

$$\hat{\tau}_{\text{FRD}} = \frac{\hat{\tau}_{Y,\text{SRD}}}{\hat{\tau}_{D,\text{SRD}}}$$

- Again, CCT provides robust bias correction, bandwidths.
 - Calonico, Cattaneo, and Titiunik (CCT; Econometrica 2014)

More Practical FRD Estimation

- The ratio estimator above is equivalent to a TSLS approach.
- Use the same specification as above with the following covariates:

$$V_i = \begin{pmatrix} 1 \\ \mathbf{1}\{X_i < c\}(X_i - c) \\ \mathbf{1}\{X_i \geq c\}(X_i - c) \end{pmatrix}$$

- First stage:

$$D_i = \delta_1' V_i + \rho \mathbf{1}\{X_i \geq c\} + v_i$$

- Second stage:

$$Y_i = \delta_2' V_i + \tau D_i + \eta_i$$

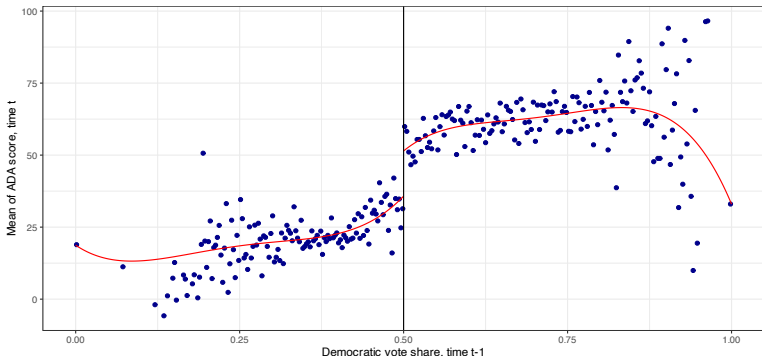
- Thus, being above the threshold is treated like an instrument, controlling for trends in X_i .

4/ R Codes for SRD and FRD

Visualization

- Visualization: showing discontinuity at the cut-off

```
1 > pacman::p_load(tidyverse, rdrobust, rddensity, ggplot2)
2 > lmb_data <- read_csv("https://bit.ly/41LCnht") ## Import data
3
4 > rdplot(y = lmb_data$score, x = lmb_data$lagdemvoteshare, ## Visualize the discontinuity
5         c = .5, title = "",
6         x.label = "Democratic vote share, time t-1",
7         y.label = "Mean of ADA score, time t")
```



Estimation

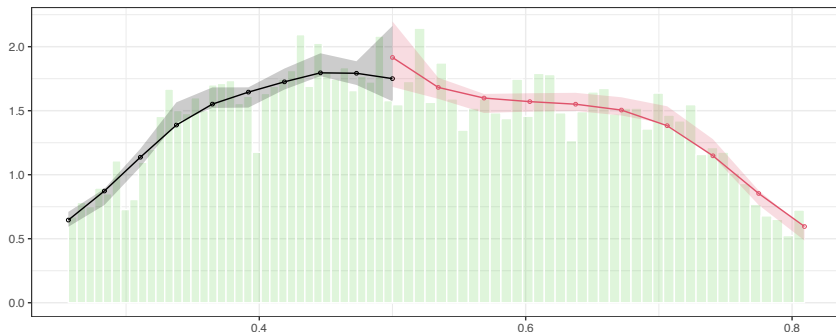
```
1 # Fit local linear regression
2 > fit <- rdrobust::rdrobust(y = lmb_data$score, x = lmb_data$lagdemvoteshare, c = .5)
3 > cbind(fit$coef, fit$se)
4
5           Coeff Std. Err.
6 Conventional  18.66595  1.692466
7 Bias-Corrected 18.44932  1.692466
8 Robust        18.44932  2.037871
```

- Two types of point estimates:
 - The standard local linear estimator, $\hat{\tau}_{\text{srd}}$
 - The local linear estimator with bias-correction, $\hat{\tau}_{\text{BC}} = \hat{\tau}_{\text{srd}} - \widehat{\text{bias}}$ (e.g., CCT 2014)
- Two standard errors:
 - Standard SE, $\hat{\sigma}^2$
 - “Robust” SE, $\hat{\sigma}_{\text{robust}}^2$, accounts for uncertainty in bias estimation.
- $\rightsquigarrow$ We report the “Robust” estimate: $\hat{\tau}_{\text{BC}}$ with $\hat{\sigma}_{\text{robust}}^2$.

Diagnostics: No sorting?

- McCrary density test:

```
1 test1 <- rddensity::rddensity(lmb_data$demvoteshare, c = .5)
2 rdplotdensity(rdd = test1,
3               X = lmb_data$demvoteshare,
4               type = "both") # lines? points? both?
```

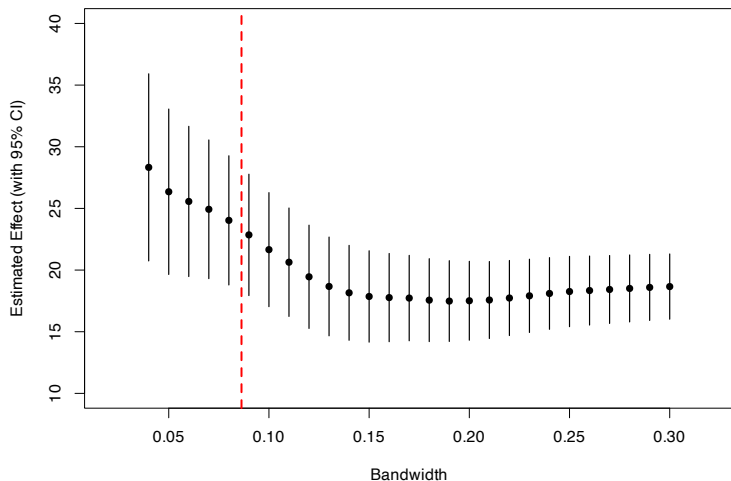


Estimated Effects Along Different Bandwidths

- Demonstrate how the results change w/ different bandwidths:

```
1 # fit local linear regression with bandwidth bws[b]
2 bws <- seq(.04, .3, by = .01); fits <- list()
3 b = 1
4 for (b in 1:length(bws)) {
5   fits[[b]] <- rdrobust(y = lmb_data$score,
6                         x = lmb_data$lagdemvotesshare,
7                         c = .5, h = bws[b])
8 }; fits
9
10 # summarize result (use "robust")
11 plot(1, 1, type = 'n', xlim = c(0.02, .32), ylim = c(10,40),
12      xlab = 'Bandwidth', ylab = 'Estimated Effect (with 95% CI)')
13
14 for (b in 1:length(bws)) {
15   points(x = bws[b], y = fits[[b]]$coef[3], pch = 16)
16   lines(c(bws[b], bws[b]), fits[[b]]$ci[3,], lwd = 1.2)
17 }
18 # abline(v = fit$bws[1,1], col = 'red', lwd = 1.5)
19 abline(v = fit$bws[1,1], col = 'red', lwd = 2, lty = "dashed")
```

Bandwidth Robustness Check



Free Tutoring Program and Academic Performance

- The Setting:
 - Students take an entrance exam at the beginning & end of a school year.
 - Those who scored below 70 are enrolled in a free tutoring program.

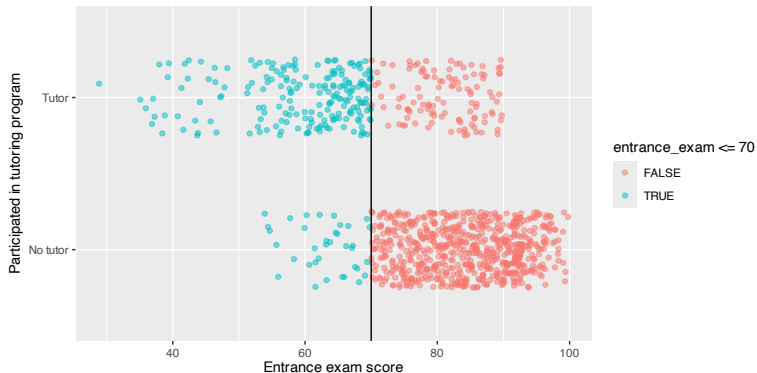
Name	Description
<code>id</code>	student id
<code>entrance_exam</code>	student's entrance exam score (out of 100)
<code>exit_exam</code>	student's exit exam score (out of 100)
<code>tutoring</code>	indicator showing if the student received tutoring

```
1 > pacman::p_load(tidyverse, broom, modelsummary, kableExtra, estimatr, rdrobust)
2 > tutoring <- read_csv("https://bit.ly/453ymbJ"); head(tutoring, 3)
3
4 # A tibble: 4 × 5
5   id entrance_exam tutoring tutoring_text exit_exam
6   <dbl>         <dbl> <lgl>      <chr>          <dbl>
7 1     1           92.4 FALSE      No tutor        78.1
8 2     2           72.8 FALSE      No tutor        58.2
9 3     3           53.7 TRUE       Tutor           62.0
```

Examine Compliance Around the Cutoff

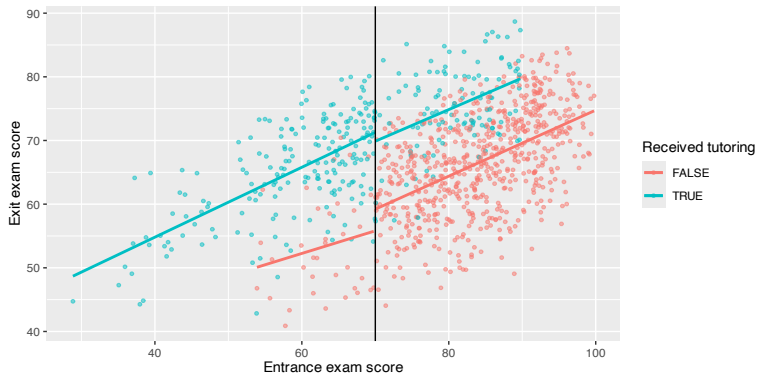
```
1 # Noncompliance around the cutoff
2 > tutoring |>
3   group_by(tutoring, entrance_exam <= 70) |>
4   summarize(count = n()) |>
5   group_by(tutoring) |>
6   mutate(prop = count / sum(count))
7
8 # A tibble: 4 × 4
9 # Groups:   tutoring [2]
10  tutoring `entrance_exam <= 70` count  prop
11  <lgl>    <lgl>                <int> <dbl>
12 1 FALSE   FALSE                 646 0.947
13 2 FALSE   TRUE                  36 0.0528
14 3 TRUE    FALSE                 116 0.365
15 4 TRUE    TRUE                  202 0.635
16
17 # Visualize Noncompliance
18 > ggplot(tutoring, aes(x = entrance_exam, y = tutoring_text, color = entrance_exam <= 70)) +
19   geom_point(size = 1.5, alpha = 0.5,
20             position = position_jitter(width = 0, height = 0.25, seed = 1234)) +
21   geom_vline(xintercept = 70) +
22   labs(x = "Entrance exam score", y = "Participated in tutoring program")
```

Visualizing Noncompliance around the Cutoff



Visualizing the Fuzzy Gap

```
1 > ggplot(tutoring, aes(x = entrance_exam, y = tutoring_text, color = entrance_exam <= 70)) +  
2   geom_point(size = 1.5, alpha = 0.5,  
3             position = position_jitter(width = 0, height = 0.25, seed = 1234)) +  
4   geom_vline(xintercept = 70) +  
5   labs(x = "Entrance exam score", y = "Participated in tutoring program")
```



Measuring the Fuzzy Gap

- Center the forcing variable, **entrance_score** and generate V_i :

```
1 > tutoring_centered <- tutoring %>%  
2   mutate(entrance_centered = entrance_exam - 70,  
3         entrance_cnt_b70 = if_else(entrance_exam < 70, entrance_centered, 0),  
4         entrance_cnt_a70 = if_else(entrance_exam >= 70, entrance_centered, 0),  
5         below_cutoff = entrance_exam <= 70)
```

- If this was a Sharp RD:

```
1 # Bandwidth ±10  
2 > sharp_parametric <- lm(exit_exam ~ entrance_centered + tutoring,  
3                          data = filter(tutoring_centered,  
4                                         entrance_centered >= -10 &  
5                                         entrance_centered <= 10))  
6 > tidy(sharp_parametric)  
7 # A tibble: 3 × 5  
8   term                estimate std.error statistic  p.value  
9   <chr>                <dbl>    <dbl>    <dbl>    <dbl>  
10 1 (Intercept)         59.3      0.503    118.    9.75e-313  
11 2 entrance_centered    0.511    0.0665     7.69  1.17e- 13  
12 3 tutoringTRUE        11.5      0.744    15.4   1.77e- 42
```

- tutoringTRUE** estimate measures the size of the jump at the cutoff.
 - There would still be confounding as we have noncompliance.

Measuring the Fuzzy Gap Correctly: TSLS

- Use `estimatr::iv_robust()` for TSLS estimation:

```
1 > model_fuzzy <- iv_robust(  
2   exit_exam ~ entrance_cnt_b70 + entrance_cnt_a70 + tutoring |  
3   entrance_cnt_b70 + entrance_cnt_a70 + below_cutoff,  
4   data = filter(tutoring_centered, entrance_centered >= -10 & entrance_centered <= 10)  
5 )  
6  
7 > tidy(model_fuzzy, conf.int = F)  
8      term      estimate std.error statistic      p.value df outcome  
9 1 (Intercept) 59.9731656 1.1165541 53.712726 9.984384e-185 399 exit_exam  
10 2 entrance_cnt_b70 0.3773301 0.1855677 2.033383 4.267520e-02 399 exit_exam  
11 3 entrance_cnt_a70 0.4715826 0.1344630 3.507156 5.044279e-04 399 exit_exam  
12 4 tutoringTRUE 9.6265535 1.9424192 4.955961 1.065759e-06 399 exit_exam
```

Measuring the Fuzzy Gap Correctly: rdrobust

- Nonparametric estimation for FRD with `rdrobust()`:

```
1 > frd <- rdrobust::rdrobust(y = tutoring$exit_exam, x = tutoring$entrance_exam,
2   c = 70, fuzzy = tutoring$tutoring, kernel = "triangular", vce = "hc2")
3
4 > summary(frd)
5
6 Fuzzy RD estimates using local polynomial regression.
7
8 Number of Obs.      1000
9 BW type          mserd
10 Kernel           Triangular
11 VCE method        HC2
12
13 Number of Obs.      238      762
14 Eff. Number of Obs. 170      347
15 Order est. (p)      1        1
16 Order bias (q)      2        2
17 BW est. (h)         12.961    12.961
18 BW bias (b)         19.579    19.579
19 rho (h/b)           0.662     0.662
20 Unique Obs.        238      762
21
22 First-stage estimates.
23
24 =====
25      Method   Coef. Std. Err.      z    P>|z|    [ 95% C.I. ]
26 -----
27 Conventional  -0.708    0.073   -9.672  0.000   [-0.851 , -0.565]
28 Robust        -      -    -8.350  0.000   [-0.909 , -0.563]
29 =====
30
31 Treatment effect estimates.
32
33 =====
34      Method   Coef. Std. Err.      z    P>|z|    [ 95% C.I. ]
35 -----
36 Conventional   9.685    1.957    4.948  0.000   [5.849 , 13.522]
37 Robust         -      -    4.127  0.000   [5.075 , 14.257]
38 =====
```

Measuring the Fuzzy Gap Correctly: Robust Estimates

- Report the “robust” estimates: $\hat{\tau}_{BC}$ with $\hat{\sigma}_{\text{robust}}^2$.

```
1 > cbind(frd$coef, frd$se)
2
3           Coeff Std. Err.
4 Conventional  9.685196  1.957433
5 Bias-Corrected 9.666085  1.957433
6 Robust        9.666085  2.342204
```



On to the Presentations & Discussions!

Contact Information:

jaewon.yoo@iss.nthu.edu.tw

<https://j1yoo.github.io/>



Appendix

Kink RD

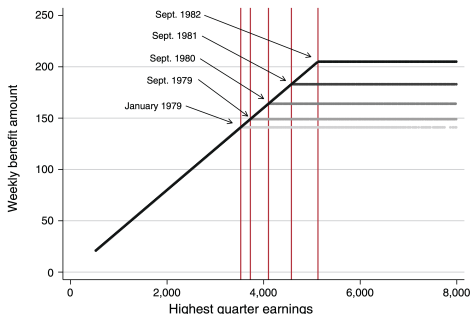


FIGURE 2. LOUISIANA: SCHEDULE OF UI WEEKLY BENEFIT AMOUNT, JAN. 1979–DEC. 1983

Source: <https://www.aeaweb.org/articles?id=10.1257/pol.20130248>

Also see: <https://blogs.worldbank.org/en/impactevaluations/tools-trade-regression-kink-design>

- **Sharp Kink RD:** discontinuities in the first derivatives rather than levels.
 - Unemployment benefits as a function of prior earnings.
 - If there is a cap on benefits, there's a kink in the assignment.
 - Look for changes in the slope of $\mathbb{E}[Y_i | X_i = x]$ at threshold.
 - Estimation similar, but better to use local quadratic regression.